

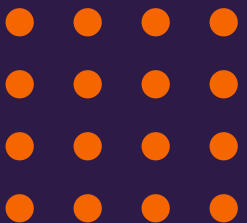


Unsupervised Learning – Clustering

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Unsupervised Learning Basics

- We want to discover subgroups among variables or observations, but not in how they related to a response variable
- There's usually no precise goal of unsupervised clustering; it's often used as a descriptive tool
- It's easier to obtain data without a response variable (i.e., label)
- Example Applications
 - Find subgroups of breast cancer patients with similar gene expressions
 - Determine subgroups of customers based on browsing and purchasing patterns
 - Group movies according to ratings assigned by viewers

Clustering

- Clustering analysis consists of techniques for finding *subgroups* or *clusters* in a multivariate data set.
- We want to partition the data into distinct groups so that observations within each group are similar to each other
- Need to define concretely what it means for observations to be *similar* or different, which can be domain specific
- Notation
 - Assume n observations are measured on p features
 - Let X be an $n \times p$ matrix of data and X_{ij} for $i = 1, 2, \dots, n$ and $j = 1, \dots, p$
 - We are interested in finding subsets of rows in X that are “similar” to each other, which will form a “cluster”



Simple Example: Eruptions of Old Faithful

Data were collected on time between eruptions and the duration of eruptions for the Old Faithful geyser at Yellowstone National Park

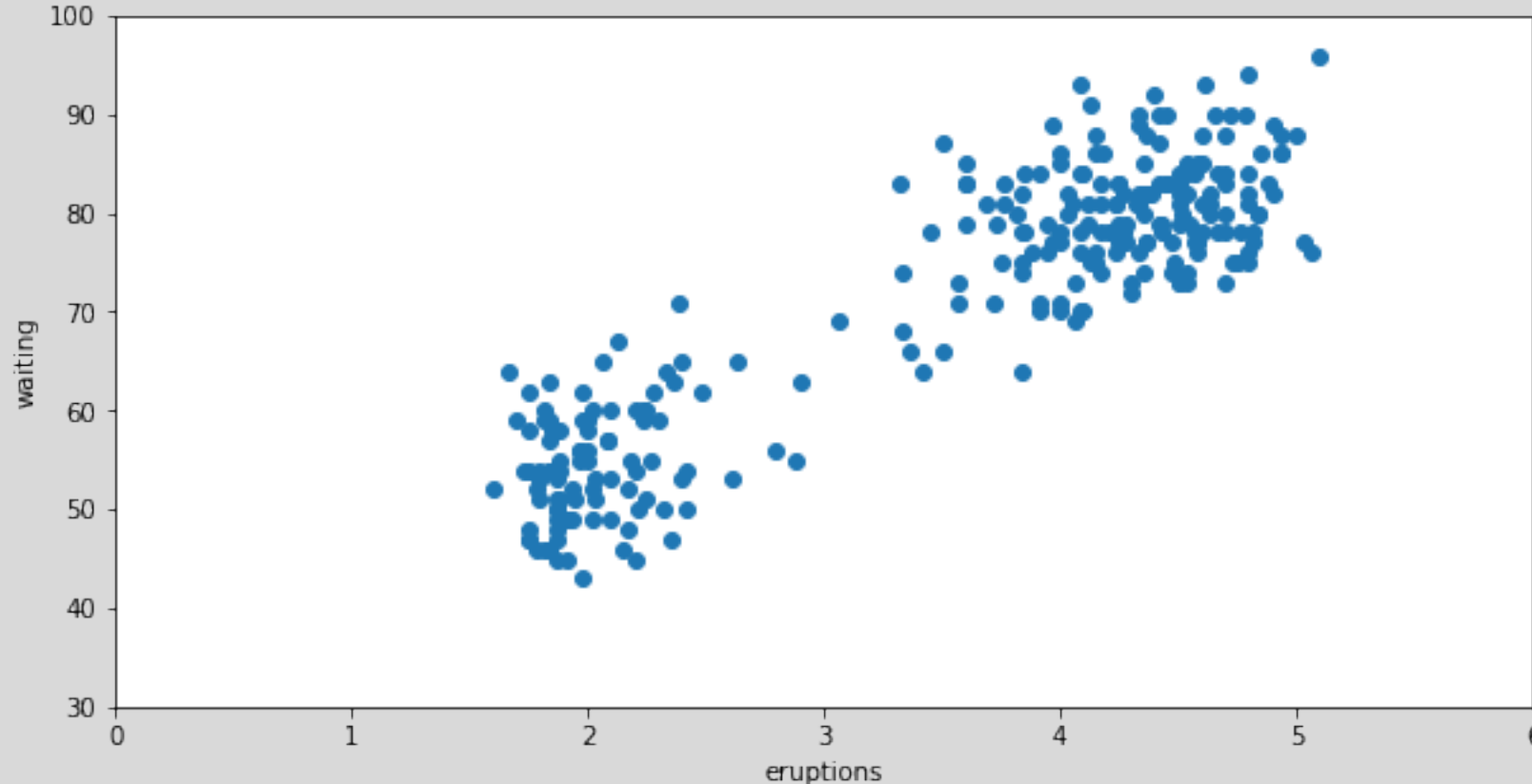
	eruptions	waiting
0	3.600	79
1	1.800	54
2	3.333	74
3	2.283	62
4	4.533	85

	eruptions	waiting
count	272.000000	272.000000
mean	3.487783	70.897059
std	1.141371	13.594974
min	1.600000	43.000000
25%	2.162750	58.000000
50%	4.000000	76.000000
75%	4.454250	82.000000
max	5.100000	96.000000



Simple Example: Eruptions of Old Faithful

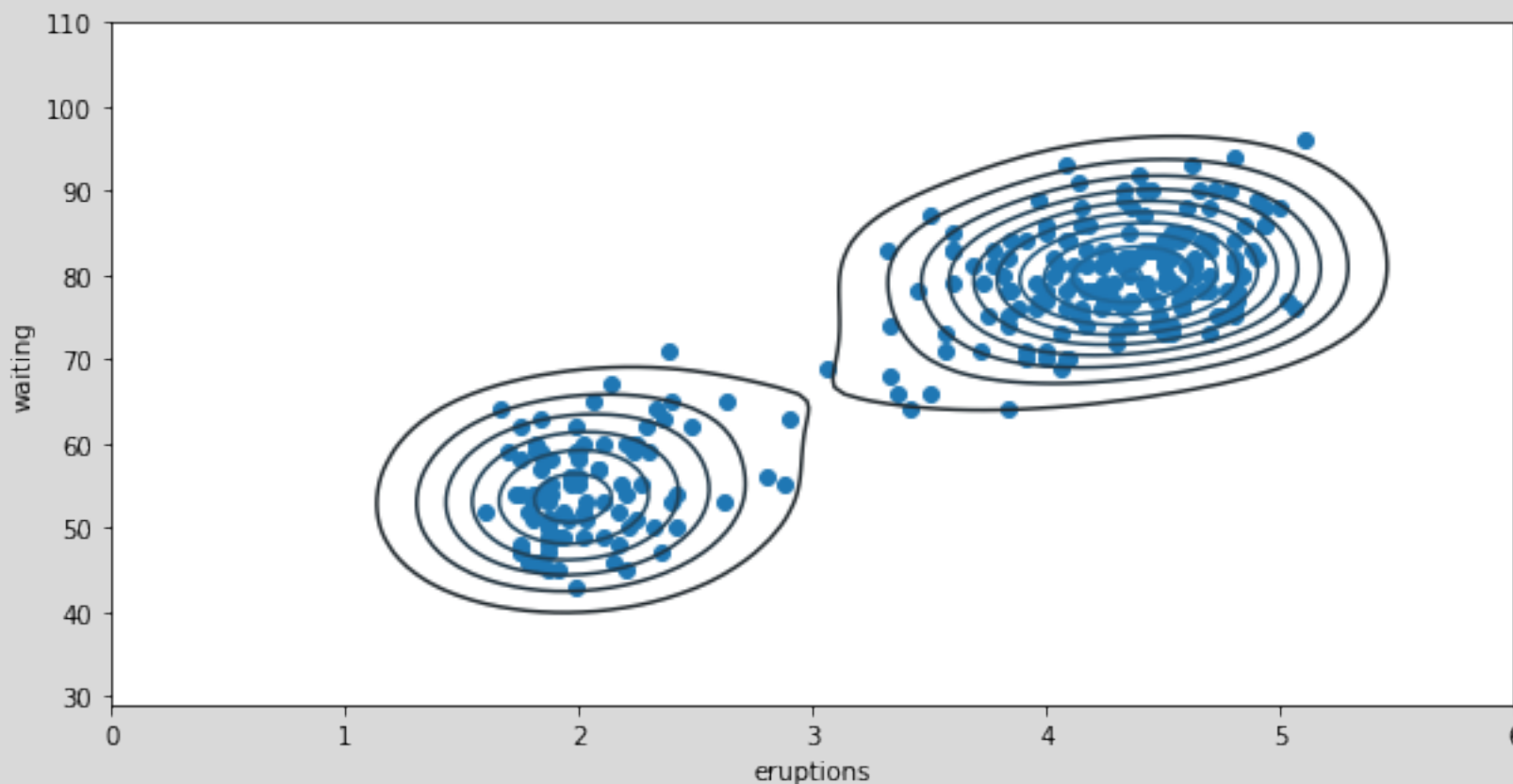
Data was collected on time between eruptions and the duration of eruptions for the Old Faithful geyser at Yellowstone National Park





Simple Example: Eruptions of Old Faithful

Data were collected on time between eruptions and the duration of eruptions for the Old Faithful geyser at Yellowstone National Park





Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering



Inter-observational Distances

- The choice of distance metrics is a critical step in clustering
- Pairwise distance calculations can influence the shape of your clusters
- Observations to be clustered can be non-standard (e.g., images, audio signals, etc.), so the process of computing distances first followed by clustering can be an effective approach.



Inter-observational Distances

- Two common distance metrics

$$d_{Euc}(x_i, x_k) = \sqrt{\sum_{j=1}^p (X_{ij} - X_{kj})^2}$$

Eucidean Distance (L_2)

$$d_{Man}(x_i, x_k) = \sum_{j=1}^p |x_{ij} - x_{kj}|$$

Manhattan Distance (L_1)

Inter-observational Distances

- Other (correlation-based) distance metrics

$$d_{Pearson}(x_i, x_k) = 1 - \frac{\sum_{j=1}^p (X_{ij} - \bar{x}_i)(X_{kj} - \bar{x}_k)}{\sqrt{\sum_{j=1}^p (X_{ij} - \bar{x}_i)^2 \sum_{j=1}^p (X_{kj} - \bar{x}_k)^2}}$$

$$d_{Spearman}(x_i, x_k) = 1 - \frac{\sum_{j=1}^p (W_{ij} - \bar{w}_i)(W_{kj} - \bar{w}_k)}{\sqrt{\sum_{j=1}^p (W_{ij} - \bar{w}_i)^2 \sum_{j=1}^p (W_{kj} - \bar{w}_k)^2}}$$

where W_{ij} are the ranks of X_{ij} for feature j , and $\bar{w}_i$ is the average of the ranks for observation i .

*Spearman correlation is used when outliers might be a concern



Distances and Scaling

Consider a data set that looks like the following:

X1	X2	X3	X4
498625	0.53	0.73	1.2
88635	0.12	0.13	2.3
623617	0.43	0.47	2.7
...	...	...	...

The distance between any two observations is basically determined by feature 1. This is an undesired consequence of variables having different scales.

Usually, we want to ensure each variable has equal contribution to the clustering algorithm, and therefore the distance computation.



Distances and Scaling

To address this problem, standardize the variables prior to computing distances

$$\frac{X_{ij} - center(x_j)}{scale(x_j)}$$

where $center(x_j)$ can be the sample mean or median of the j^{th} variable, and $scale(x_j)$ can be the sample standard deviation or mean absolute deviation of the j^{th} variable.



Example: Violent Crimes by State

This data set reports arrests per 100,000 residents for assault, murder, and rape in each of the 50 US states in 1973. Also given is the percentage of population living in urban areas.

	Murder	Assault	UrbanPop	Rape
count	50.000000	50.000000	50.000000	50.000000
mean	7.78800	170.760000	65.540000	21.232000
std	4.35551	83.337661	14.474763	9.366385
min	0.80000	45.000000	32.000000	7.300000
25%	4.07500	109.000000	54.500000	15.075000
50%	7.25000	159.000000	66.000000	20.100000
75%	11.25000	249.000000	77.750000	26.175000
max	17.40000	337.000000	91.000000	46.000000



Example: Violent Crimes by State

First few randomly chosen observations

	State	Murder	Assault	UrbanPop	Rape	StateAbbrev
0	Alabama	13.2	236	58	21.2	AL
1	Alaska	10.0	263	48	44.5	AK
2	Arizona	8.1	294	80	31.0	AZ
3	Arkansas	8.8	190	50	19.5	AR
4	California	9.0	276	91	40.6	CA



Example: Violent Crimes by State

First few observations rescaled (subtract mean, divide by standard deviation)

	Murder	Assault	UrbanPop	Rape
State				
Hawaii	-0.58	-1.51	1.22	-0.11
Indiana	-0.14	-0.70	-0.04	-0.03
New Mexico	0.84	1.38	0.31	1.17
Washington	-0.88	-0.31	0.52	0.54
Maine	-1.32	-1.06	-1.01	-1.45
Alabama	1.26	0.79	-0.53	-0.00



Example: Violent Crimes by State

Euclidean distances calculated on scaled observations

	Hawaii	Indiana	New Mexico	Washington	Maine	Alabama
Hawaii	0.000000	1.561769	3.586656	1.560979	2.743631	3.422932
Indiana	1.561769	0.000000	2.617305	1.152154	2.124266	2.097219
New Mexico	3.586656	2.617305	0.000000	2.504780	4.390177	1.615635
Washington	1.560979	1.152154	2.504780	0.000000	2.655948	2.675068
Maine	2.743631	2.124266	4.390177	2.655948	0.000000	3.520494
Alabama	3.422932	2.097219	1.615635	2.675068	3.520494	0.000000



Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering



Partition-based Clustering

Basic Idea: Specify the number of clusters into which the data will be partitioned, and then perform computation to group data so that

1. observations within clusters are similar (low distances), and
2. observations in different clusters are dissimilar (high distances).

We will address the optimal number of clusters separately.

K-Means Clustering

Let $C_1, \dots, C_K$ denote sets containing the indices of observations within the K clusters such that

1. $C_1 \cup C_2 \cup \dots \cup C_K = \{1, 2, \dots, n\}$, and
2. $C_k \cap C_l = \emptyset$ for $k \neq l$.

Let $W(C_K) = \frac{1}{|C_K|} \sum_{i, i' \in C_K} \sum_{j=1}^p (X_{ij} - X_{i'j})^2$ be the within cluster variation.

The *goal of K-Means clustering* is to find $C_1, \dots, C_K$ such that $\sum_{k=1}^K W(C_k)$ is minimized.

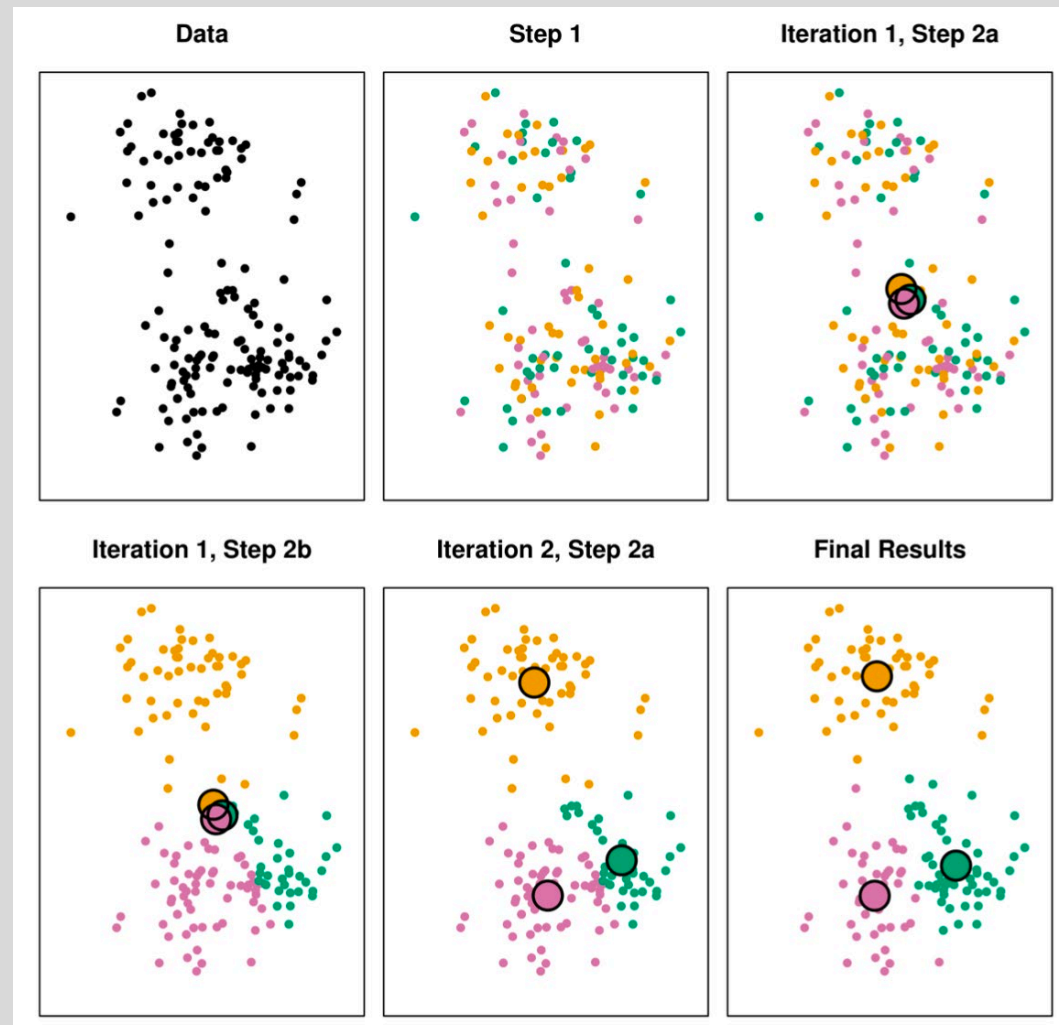


K-Means Clustering Algorithm

1. Randomly assign each observation to one of K clusters at random.
2. Repeat the following steps until clusters do not change:
 1. For each cluster k , compute the cluster centroid $\bar{x}_k$ (the variable-wise average of the observations in cluster k).
 2. Given the k centroids, reassign all observations to clusters based on their closeness to the centroids.



K-Means Clustering Algorithm





K-Means Clustering Algorithm

- Requires you to select K in advance.
- The algorithm is *locally optimal*, not globally optimal.
 - i.e., you can get different cluster results depending on the initial cluster assignments.
- Potential Solutions
 - Try different values of K and compare the results.
 - Try different initial cluster assignments in parallel and chose the one with the best within-cluster sum of squared deviations.

K-Means Clustering Algorithm

- Six results from different initializations





Example: Violent Crimes by State

First few observations rescaled (subtract mean, divide by standard deviation)

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State				
Hawaii	-0.58	-1.51	1.22	-0.11
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Alabama	1.26	0.79	-0.53	-0.00



Example: Violent Crimes by State

We chose 4 clusters for now

	Murder	Assault	UrbanPop	Rape	Cluster
State					
Alabama	1.255179	0.790787	-0.526195	-0.003451	1
Alaska	0.513019	1.118060	-1.224067	2.509424	1
Arizona	0.072361	1.493817	1.009122	1.053466	1
Arkansas	0.234708	0.233212	-1.084492	-0.186794	2
California	0.281093	1.275635	1.776781	2.088814	1

Clustering Diagnostics: Silhouette Plot

Once a clustering has been determined, let

- a_i = average dissimilarity between observation i and the other points in the cluster to which i belongs
- b_i = average dissimilarity between observation i and the other points in the next closest cluster to observation i .

We define $s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$ to be the silhouette score for observation i .

Clustering Diagnostics: Silhouette Plot

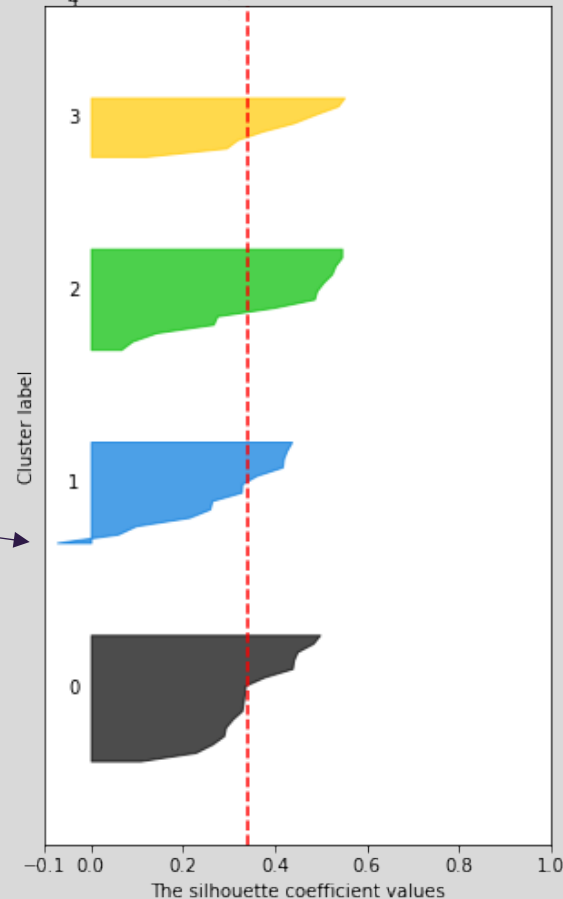
$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

- Interpretation
 - Observations with $s_i \approx 1$ are well-clustered
 - Observations with $s_i \approx 0$ lie between two clusters
 - Observations with $s_i < 0$ are probably in the wrong cluster

Example: Violent Crimes by State

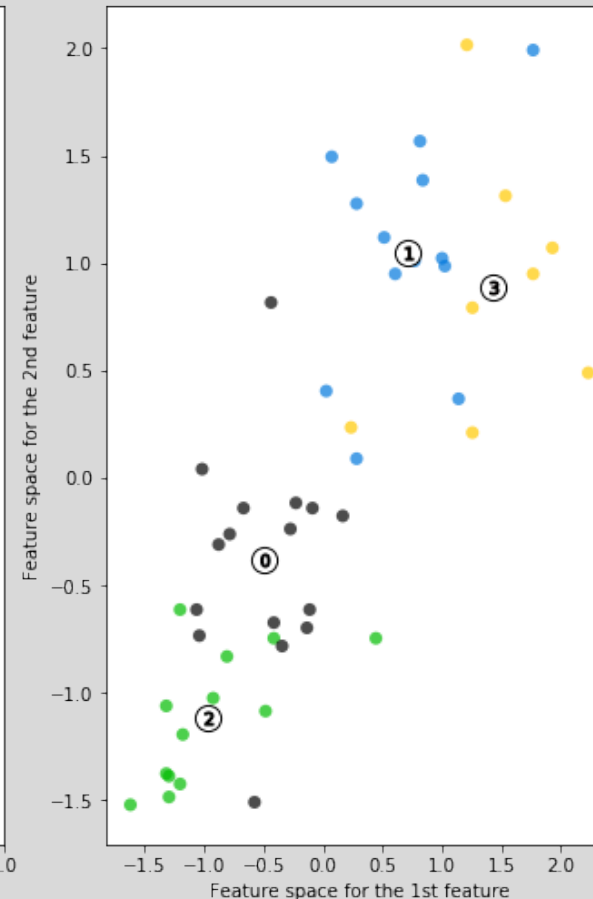
Silhouette analysis for KMeans clustering on sample data with $n_clusters = 4$

The silhouette plot for the various clusters.



Something
may be in the
wrong cluster

The visualization of the clustered data.





Example: Violent Crimes by State

	State	Murder	Assault	UrbanPop	Rape	StateAbbvr
24	Missouri	9.0	178	70	28.2	MO



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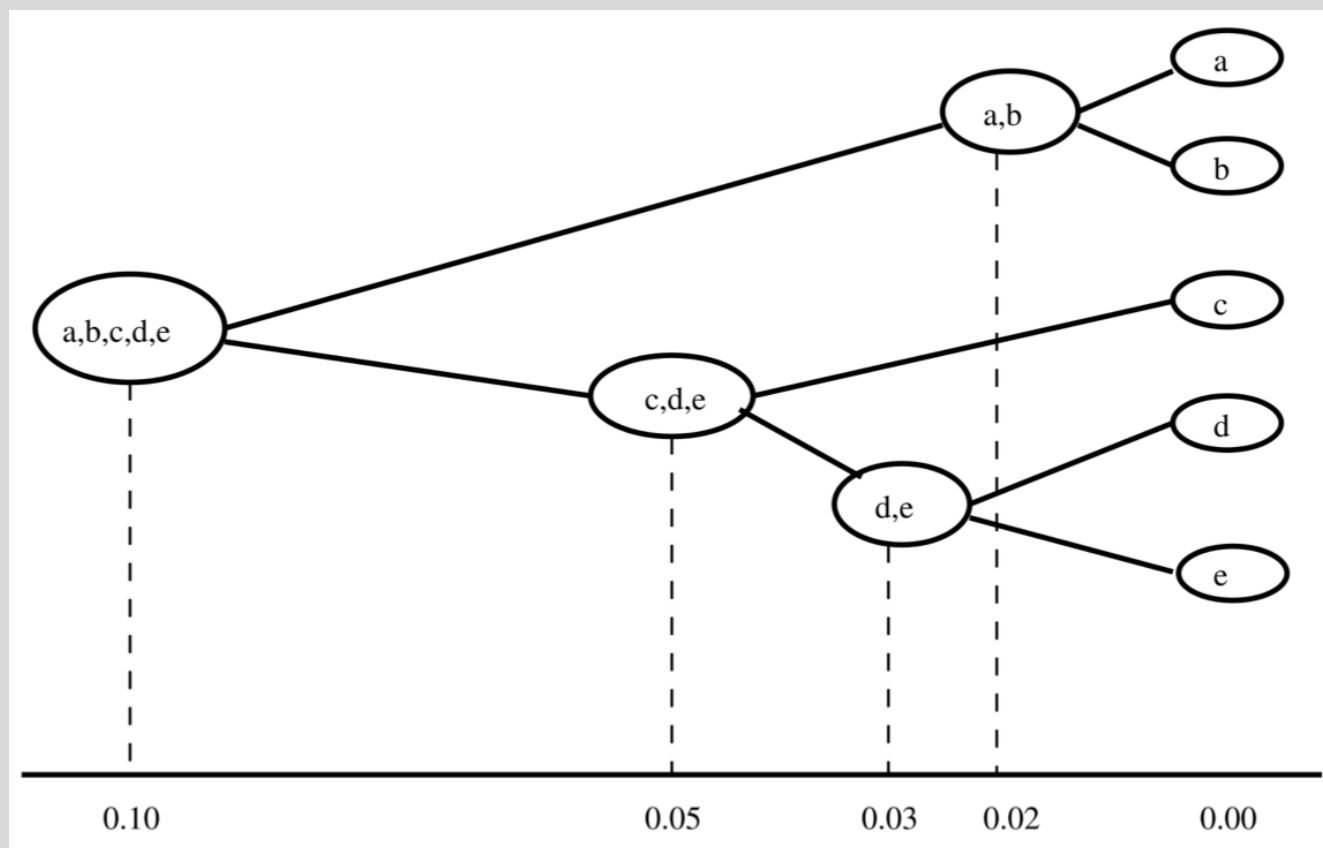
Hierarchical Clustering

- K-Means clustering requires pre-specifying the number of clusters
- Hierarchical clustering does not require committing to a particular number of clusters
- Two types of hierarchical clustering
 - Agglomerative clustering (bottom-up approach)
 - Divisive clustering (top-down approach)



Hierarchical Clustering

Hierarchical clustering can be summarized in a dendrogram





Agglomerative Clustering

The basic algorithm:

- Each observation starts as its own cluster
- At each step of the algorithm, two clusters that are most “similar” are combined into a new larger cluster
- This process of combining clusters is repeated until all observations are members of one single large cluster

Agglomerative clustering is particularly well-suited for identifying small clusters.

Agglomerative Clustering

A few common approaches:

- Complete (or maximum) Linkage Clustering: For two clusters, determine the maximum dissimilarity between any observation in the first cluster and any observation in the second cluster.
- Single Linkage Clustering: For two clusters, determine the minimum dissimilarity between any observation in the first cluster and any observation in the second cluster.
- Average Linkage Clustering: Compute all pairwise dissimilarities between observations in the first and second cluster and calculate the average.

Other aggregation approaches are also possible.



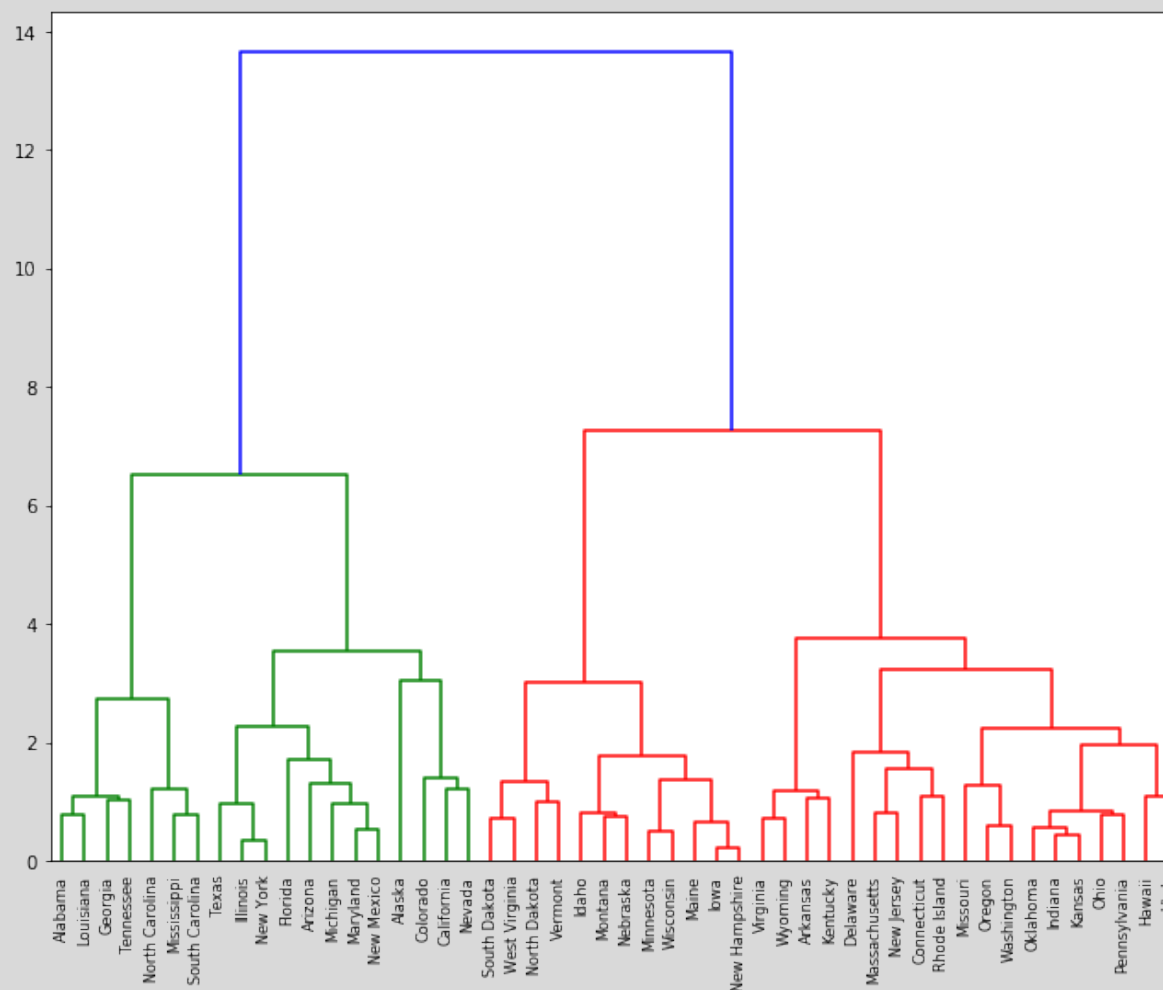
Agglomerative Clustering

A popular approach is known as Ward's Method:

- Instead of joining two clusters based on their distances, use information about the variance of observations within clusters.
- Specifically, at each step, join two clusters whose merged cluster has the smallest within-cluster sum of squared distances.



Agglomerative Clustering – Ward's Method





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Optimal Cluster Counts

But... how do we choose the optimal number of clusters?

There is no single compelling way to choose clusters, but we will explore two types of methods:

- **Direct Methods** that involve optimizing a particular criterion
 - Silhouette method
 - Elbow method
- **Testing Methods** that evaluate evidence against a null hypothesis
 - Gap Statistic

Elbow Method

As previously, let $W(C_k)$ be the within-cluster sum of squared distances between all pairs for cluster k for a particular clustering, and let

$$T_K = \sum_{k=1}^K W(C_k)$$

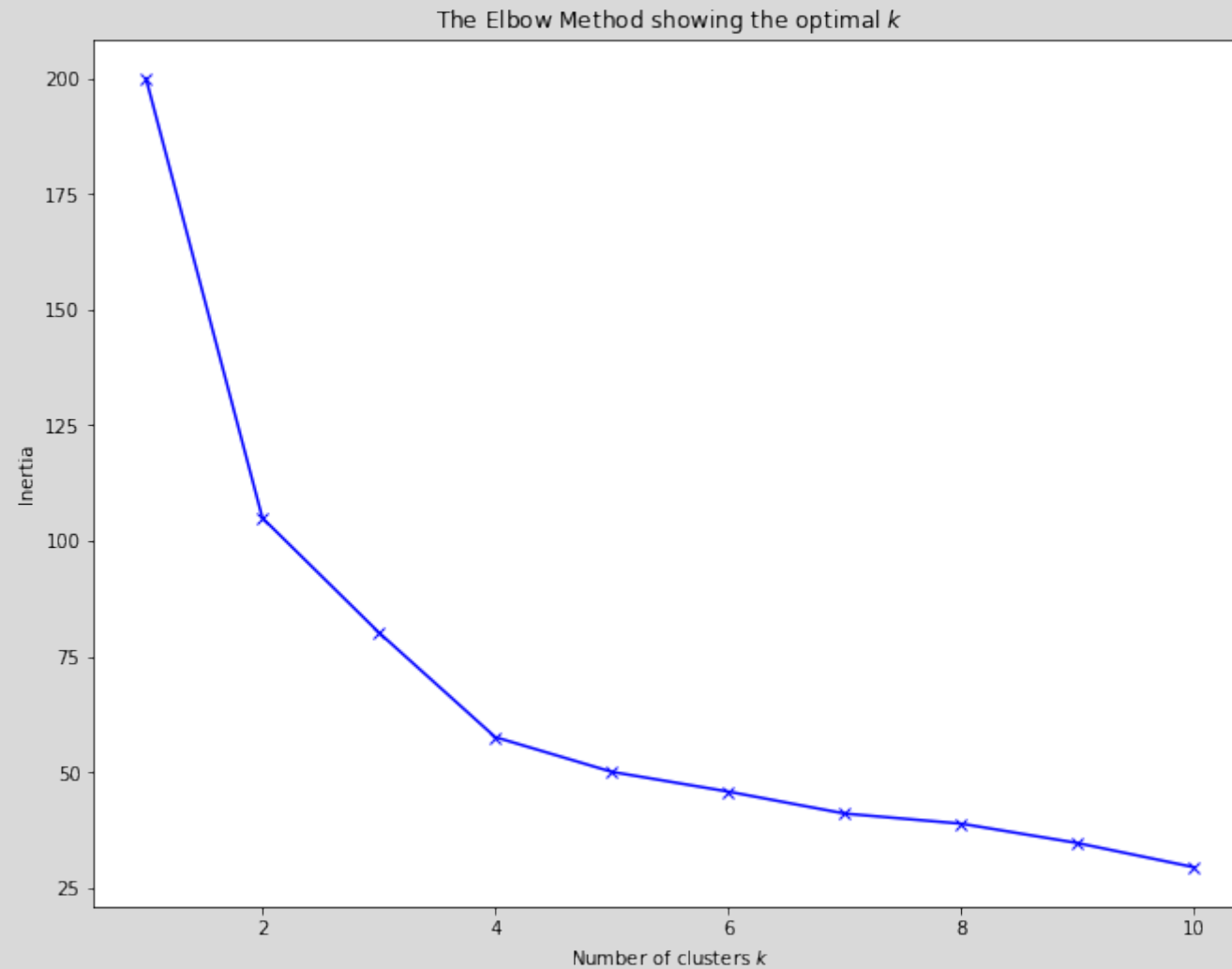
be the total within-cluster variation for the clustering of K clusters.

1. For the particular clustering method, let K vary over a range of values (e.g., 1 to 10)
2. Compute T_K for each K
3. Plot T_K against K and look for a clear bend in the graph

The value where the bend occurs is considered the correct number of clusters



Elbow Method



Average Silhouette Method

Similar to the elbow method:

1. For the particular clustering method, let K vary over a range of values (e.g., 1 to 10)
2. For each K , calculate the average silhouette across all observations

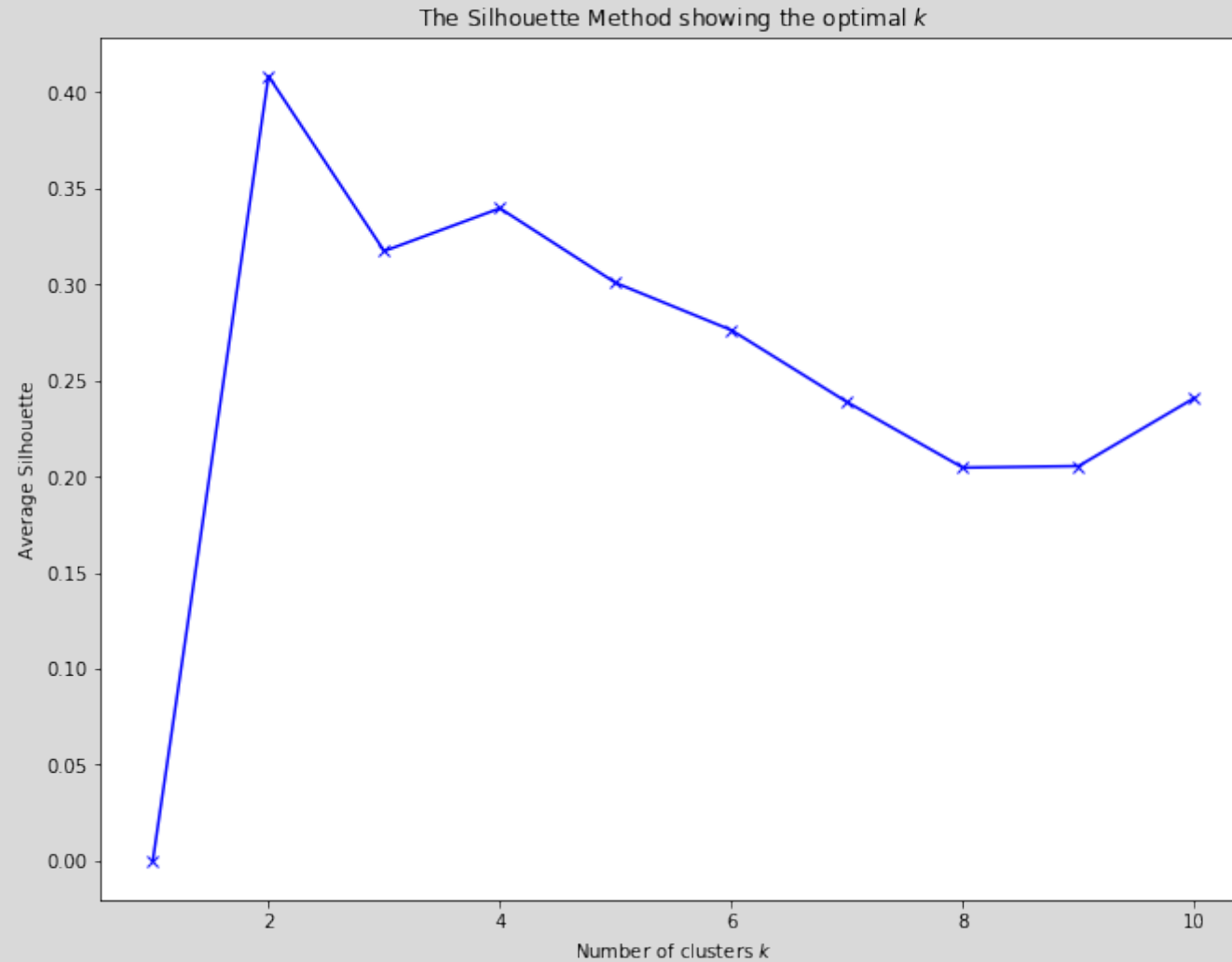
$$S_K = \frac{1}{n} \sum_{i=1}^n s_i$$

3. Plot S_K against K

The value of K where S_K is maximized is considered the appropriate number of clusters.



Average Silhouette Method





Comments

- The elbow method suggests four clusters for K-Means
- Average silhouette method suggests two
- Computation for each method are different enough that inconsistent results are not uncommon
- Both approaches measure global clustering characteristics only, and are informal (and somewhat ad hoc approaches)

Gap Statistic

- **Idea:** For a particular choice of K clusters, compare the total within-cluster variation to the expected within-cluster under the assumption that the data have no obvious clustering (i.e., randomly distributed)
- The Gap Statistic in essence detects whether the data clustered into K groups is significantly better than if they were generated at random

Gap Statistic Algorithm

1. Cluster the data at varying number of total clusters K . Let T_K be the total within-cluster sum of squared distances
2. Generate B reference data sets of size n , with the simulated values of variable j uniformly generated over the range of the observed variable x_j . Typically, $B = 500$.
3. For each generated data set $b = 1, \dots, B$, perform the clustering for each K . Compute the total within-cluster sum of squared distances, $T_K^{(b)}$.
4. Compute the Gap Statistic: $Gap(K) = (\frac{1}{B} \sum_{b=1}^B \log(T_K^{(b)})) - \log(T_K)$
5. Let $\bar{w} = \frac{1}{B} \sum_{b=1}^B \log(T_K^{(b)})$. Compute the standard deviation
 $sd(K) = \sqrt{\frac{1}{B} \sum_{b=1}^B (\log(T_K^{(b)}) - \bar{w})^2}$. Define $s_K = sd(K) \sqrt{1 + 1/B}$.
6. Finally, **choose the number of clusters as the smallest K such that**
$$Gap(K) \geq Gap(K + 1) - s_{K+1}$$

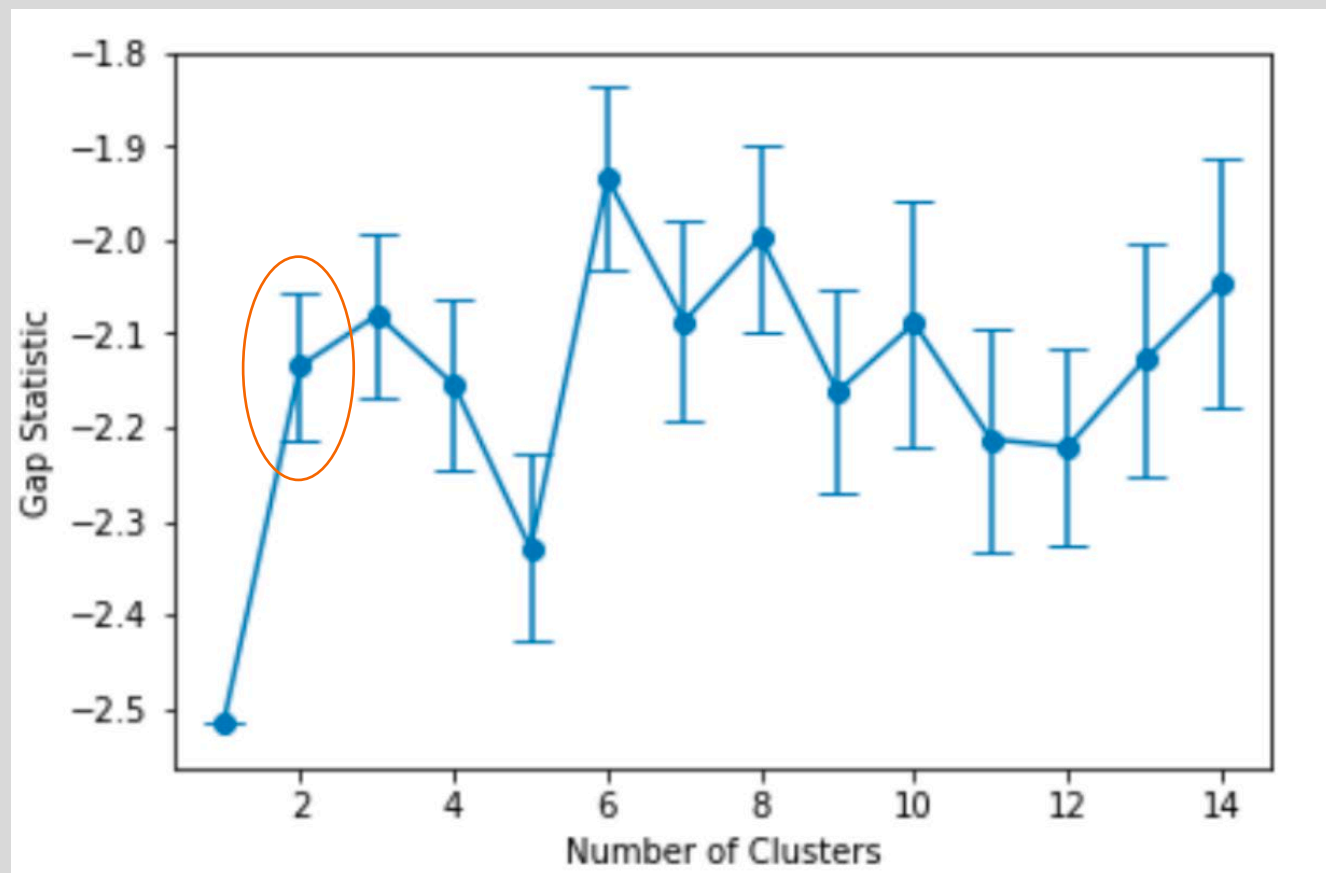


Gap Statistic Algorithm

	n_clusters	gap_value	gap*	ref_dispersion_std	diff	diff*
0	1.0	-2.518049	-183.840402	1.082720	-0.327729	1.418971
1	2.0	-2.108007	-92.168043	1.061159	0.042001	48.885993
2	3.0	-2.063404	-70.175014	0.892529	-0.011188	29.238100
3	4.0	-1.953498	-49.457242	0.798298	0.164137	42.875510
4	5.0	-2.013890	-45.934873	0.734503	0.098662	35.904791



Gap Statistic Algorithm



Gap Statistic Algorithm

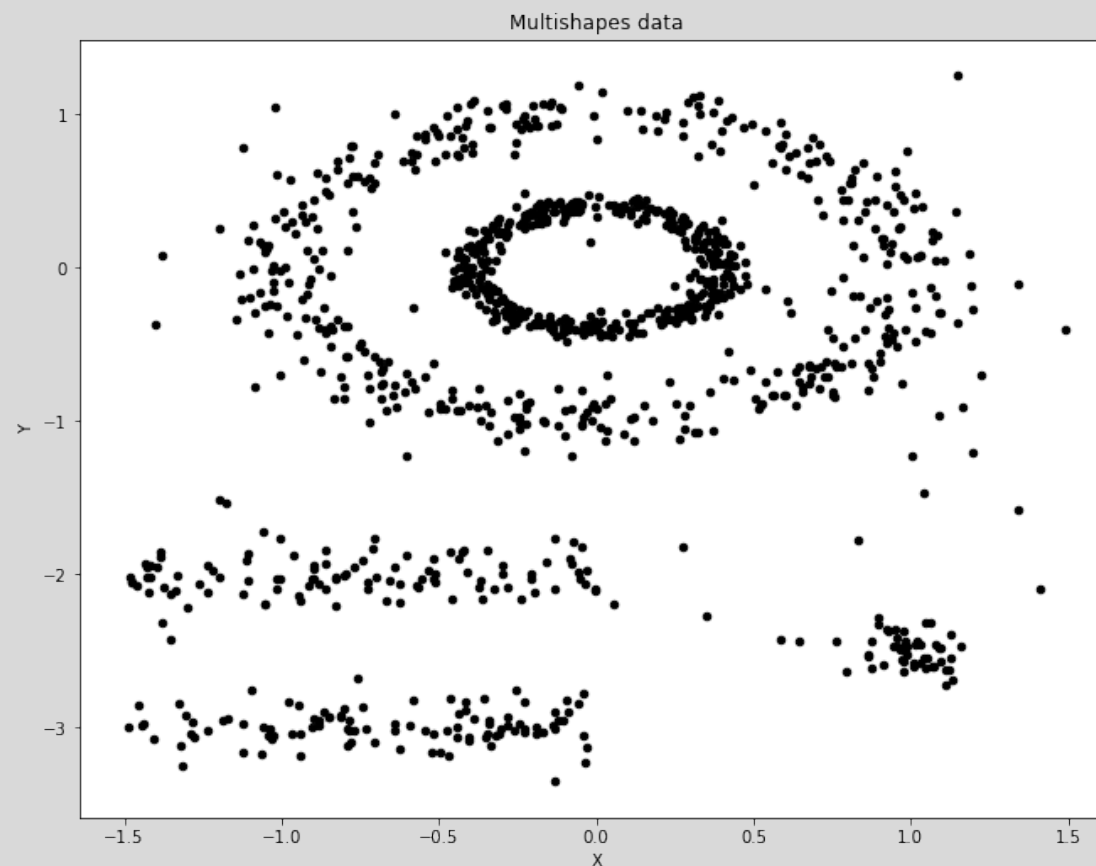
- K-Means clustering is optimized for $K = 2$ based on the Gap Statistic
- This is a principled approach to choosing cluster sizes, though clearly different conclusions can be reached based on different clustering approaches
- The Gap Statistic is generally understood to be conservative—tends to err on the side of picking a smaller number of clusters
- There's an alternative form of the gap statistic which replaces the $\log(T_K)$ and $\log(T_K^{(b)})$ terms in the formula with

$$Gap^*(K) == \left(\frac{1}{B} \sum_{b=1}^B T_K^{(b)} \right) - T_K$$

This turns out to be less conservative.

More Clustering

- Imagine trying to cluster something like this

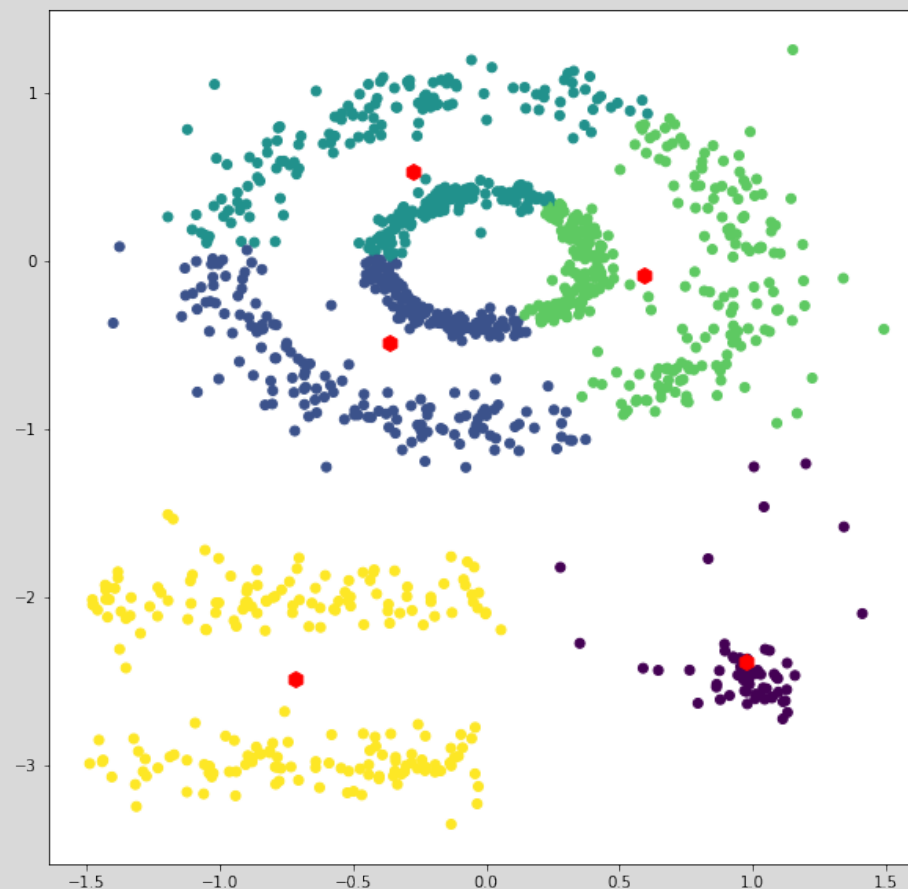


What happens when we try K-means?



More Clustering

- K-Means struggles with this data set.





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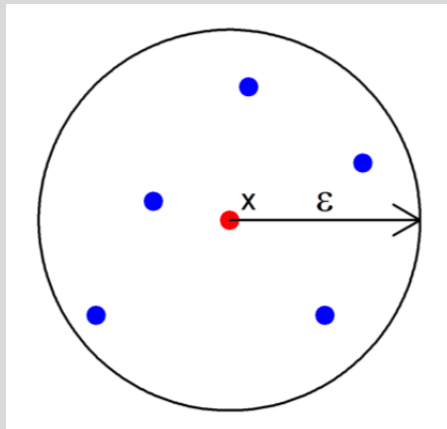
DBSCAN: Density-based Clustering Algorithm

- Unlike previous clustering algorithms, DBSCAN
 - Can find any shape of clusters
 - Identifies observations that do not belong to clusters as outliers
 - Does not require specifying the number of clusters
 - Can be used for predicting cluster membership for new data

DBSCAN: Density-based Clustering Algorithm

The DBSCAN algorithm identifies dense regions of observations. However, we need to specify two parameters for the algorithm:

1. ϵ : the radius of a neighborhood around an observation
2. *MinPts*: the minimum number of points within an ϵ radius of an observation to be considered a “core” point.



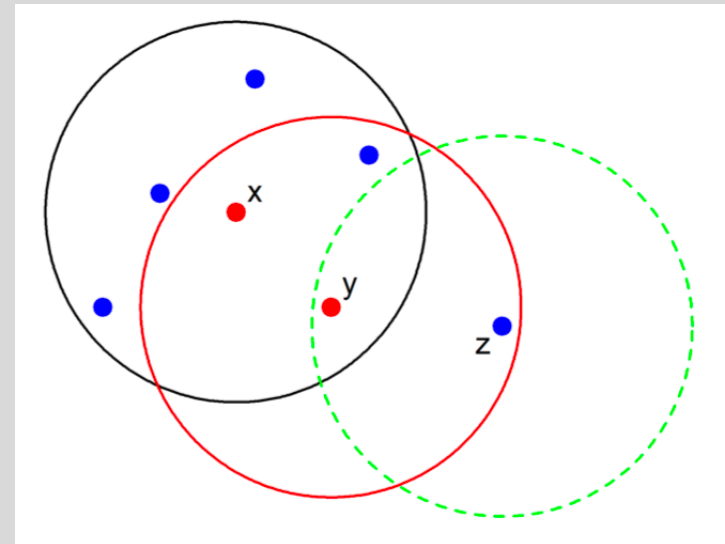
Example of core observation
with $MinPts = 6$

DBSCAN: Density-based Clustering Algorithm

There are three types of points:

- **Core points:** observations with $MinPts$ total observations within an ϵ radius
- **Border points:** observations that are not core points, but are within ϵ of a core point
- **Noise points:** everything else

In this example, x is a core point, y is a border point, and z is a noise point.



DBSCAN: Density-based Clustering Algorithm

Two terms:

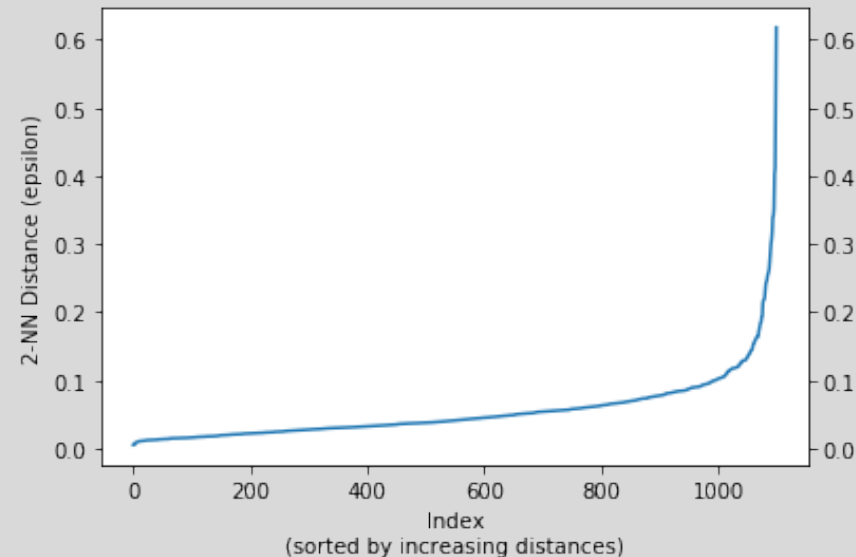
- **Density-reachable:** Point A is density-reachable from point B if there is a set of core points leading from B to A .
- **Density-connected:** Two points A and B are density-connected if there is a core point C such that both A and B are density-reachable from C .

Therefore, a density-based cluster is defined as a group of density-connected points.

DBSCAN: Density-based Clustering Algorithm

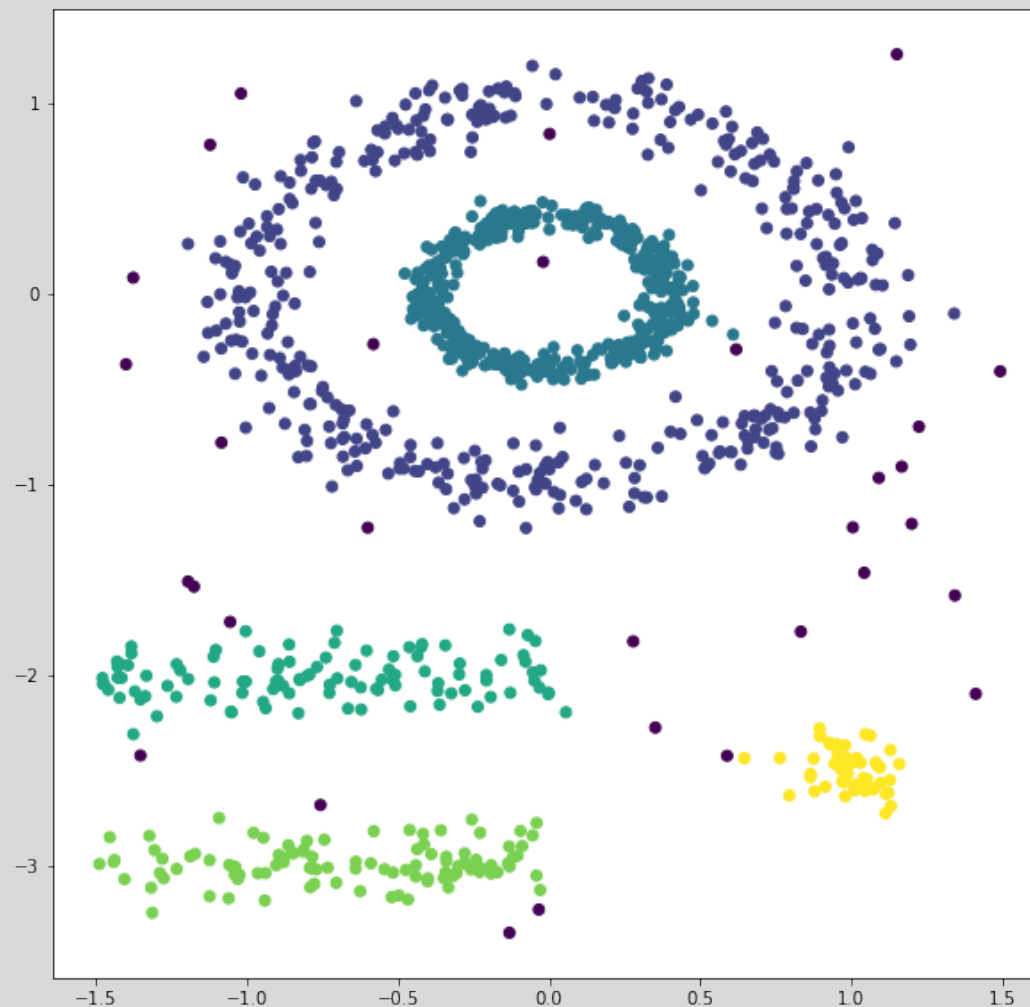
Choosing ϵ :

- Compute the k -th nearest neighbor distance for each point
- Plot the distances in sorted order
- Look for a bend (the “knee”) in the plot and use the distance at the knee as the choice of ϵ .





DBSCAN: Density-based Clustering Algorithm





Questions?